Fault Atlas

Cryptography

```
❯ grep -B 2 -A 8 "if rnd == 9" model.py

        # Fault injection hook in Round 9 right before MixColumns
        if rnd == 9 and fault is not None:
            c, r, v = fault
            state[4 * c + r] ^= v

        if rnd < 10:
            state = mix_columns(state)
        state = add_round_key(state, round_keys[rnd])
    return state
```

I began by reading `model.py` and `data.json`. The challenge performed Differential Fault Analysis (DFA) on AES-128, injecting random single-byte faults into the State right before Round 9 MixColumns.

I n `data.json`, 110 pairs of (ciphertext, faulty_ciphertext) were provided along with the corresponding fault coordinates (column c, row r, and fault value v).

Because the round 9 MixColumns layer diffuses a single faulted byte across an entire 4-byte column in round 10, each pair provided 3 nonlinear filtering equations on the 4 round-10 key bytes. Intersecting hypothesis spaces recovered the round-10 key and inverted the AES key schedule.

Finally, I tested the recovered flag against the executable / service:

`python3 Fault_Atlas.py`

Master Key recovered: ef06894cdda41099d17786e51911ce45

Flag Captured

`SecLeaf{f4ult5_d0nt_n33d_t0_b3_h0n35t}`

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